Group and solo flight: comparisons

What differs between group time and solo flight, and what remains when the controls are in place?

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Abstract

Group time differs from solo flight, but so do the pilots, the places, the routes and the shape of the flights. This report compares paragliders in three populations, solo flights, group time and the same flights alone, and gives for each result the number of flights or pieces behind it. Flights declared open fold back on themselves, and their shape enters the mean squared displacement: we keep those whose landing is at least 0.5 of the path flown from the takeoff. On them the three populations progress almost alike, $H_{\mathrm{loc}}(1000\ \mathrm{s})$ of 0.947 in group time against 0.940 in solo flights, up to lags of hours; group time is straighter only in the larger groups, and it climbs more and glides faster and flatter. Against the same route flown on another day by isolated solo flights of the same wing level, on the declared open flights, the gap $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$ falls from $+0.061$ to $+0.016$ [$+0.004$, $+0.028$] on 3 224 pieces: the route explains 74% of it (78% for experienced pilots, 63% for beginners). What remains is small, and whether the group causes it stays open.

1 Aim

Does a paraglider move differently in a group than alone? This report answers with a series of comparisons of group time with solo flight. Each has its own control, and a result is read together with the number of flights or pieces behind it, given in every table and figure.

All the populations are paragliders. Section 2 defines them, Section 3 shows that the flights declared open fold back on themselves and keeps the straight ones, counts them and maps them, Section 4 compares the populations as they are, over lags up to hours and by the size of the group, Section 5 repeats the comparison in the zones with the most group flights, and Section 6 compares group time with the same route flown alone.

2 Populations and measures

Which flights, which stretches of flight, and how many of them?

Groups. The labels are those of the note *Group flights in free flight* (`methods/group-identification`). Two pilots are in contact when one is on the path of the other within $\delta = 2$ min, $\varepsilon_h = 500$ m horizontally and $\varepsilon_z = 300$ m vertically (headings compared only in glide) for at least $\tau^* = 38.1$ min. The configurations of an instant are the connected components of the contact graph, and a group keeps the same members over a maximal interval. A contact means shared air within sight; it does not prove interaction.

States. After its *departure*, the first fix 3 km from the takeoff, every instant of a flight is in a group or solo; before it the pilot counts as neither. Flights that never depart are left out. We use paraglider flights logged every 10 s or finer, so that the labels exist at every instant; the phases (climb, search, glide) exist on the flights logged every second.

Populations. Three, all of paragliders.

- S, *solo flights*: flights that never entered a group, the whole of their time after the departure. They may have had contacts shorter than τ^*.
- G, *group time*: the stretches in a group of the flights that entered one at least once. A stretch stays in one group; G counts a flight once however many groups it joins.

- A, *the same flights alone*: the solo stretches of those same flights, a control with the same pilots, gliders, loggers and days as G.

The flights are the open ones of Section 3. The control of Section 6 is stricter than S and is defined there.

Measure. A *piece* is a stretch within one segment of the track, of at least 1250 s, on a 10-s grid of its clock. For a flight f and a population, the time-averaged mean squared displacement pools the origins of all its pieces in that population,

$$\overline{\delta^2_f}(\tau) = \frac{1}{N_f(\tau)} \sum_t \|\mathbf{r}(t+\tau) - \mathbf{r}(t)\|^2, \tag{1}$$

with t and $t+\tau$ in the same piece. The ensemble MSD(τ) is its mean over the flights, one weight per flight, and H is half the slope of log MSD against log τ over the lags from 10 to 1000 s: $H = 1$ for a straight line, $H = \frac{1}{2}$ for a random walk. $H_{\text{loc}}(\tau)$ is the same slope over the lags within ± 0.25 decades of τ. It is read at $\tau = 1000$ s, over one to two climb–glide cycles. Climbs and glides are summarised by the share of time in climb, the share of searches that end in a climb, the climb rate, the glide speed and the glide ratio, each a ratio of sums over the phase runs of a population. Bands are 5–95% of bootstrap draws that resample the pilots and the dependency blocks of the grouping.

3 Open flights

Which flights do we compare, and why a criterion on their shape?

Declared open. We start from the flights the pilot declared open on the FFVL catalogue: free distance and distance with 1, 2 or 3 turn points; triangles, quadrilaterals and out-and-returns are closed. The declared type agrees with the track (takeoff and landing more or less than 6 km apart) for 91% of the 154 598 flights. 66 845 of them are declared open.

They fold. A declared open flight is not a straight line. Its *opening*, the distance from takeoff to landing over the path flown, has a median of 0.32, and of 0.23 for the 14 056 flights of 4 h or more, the ones the longest lags rest on (Figure 1, left). At each turn point, and along each ridge flown back, the pilot changes direction: over lags longer than a leg the displacement grows less than the time, and H_{loc} falls. On the declared open flights it is 0.827 in S, 0.895 in G and 0.808 in A at 11 450 s (Figure 1, right; long series on 5 775, 158 and 853 flights).

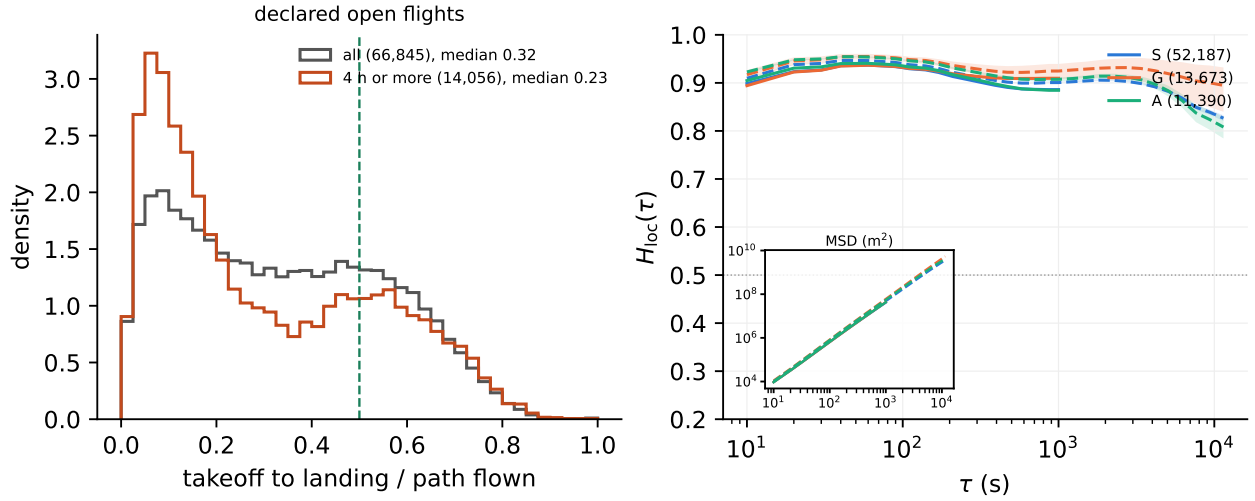


Figure 1: Declared open flights. Left: their opening, takeoff to landing over the path flown, all of them and those of 4 h or more; dashed, the threshold 0.5. Right: $H_{\text{loc}}(\tau)$ of S, G and A with an inset of MSD(τ); thin, all the pieces up to 1000 s, with the flights in the legend; dashed, the pieces long enough for 11 450 s.

The criterion. We study the difference between group and solo flight, not the shape of the flights. The MSD and H measure displacement, so if group and solo flights fold differently they differ through their shape, whatever the group does; the median opening is 0.28 for the declared open flights with group time and 0.33 for the others. Without a criterion on the shape the two effects are confounded. We keep the declared open

flights with an opening of at least 0.5: 17 687 flights, 26% of the declared open ones and 24% of those of 4 h or more. The criterion is needed for the MSD and H; the statistics of the phases, such as the glide ratio or the share of searches that end in a climb, do not depend on the shape of the whole flight. We keep one population for every statistic, so that all the comparisons rest on the same flights. From here on, *open* means declared open with an opening of at least 0.5.

	solo flights S	group time G	same flights alone A
flights, H and H_{loc}	14 192	3 342	2 577
pieces of at least 1250 s	14 970	4 446	3 738
flights, time in climb and searches	8 158	2 432	2 276
flights, climb rate and glides	12 945	3 346	2 850

Table 1: Members of the populations, on the straight open flights. The flights of G and A are the same flights, and A keeps those with a solo stretch of at least 1250 s.

Sizes. Table 1 counts the members. G has 3 342 flights in 4 446 pieces and S has 14 192 flights in 14 970. A has 2 577 flights in 3 738 pieces, fewer flights than G: the others have no solo stretch long enough. The phase statistics rest on fewer flights, those logged every second.

Where. Figure 2 maps the middle of the pieces of S and of G.

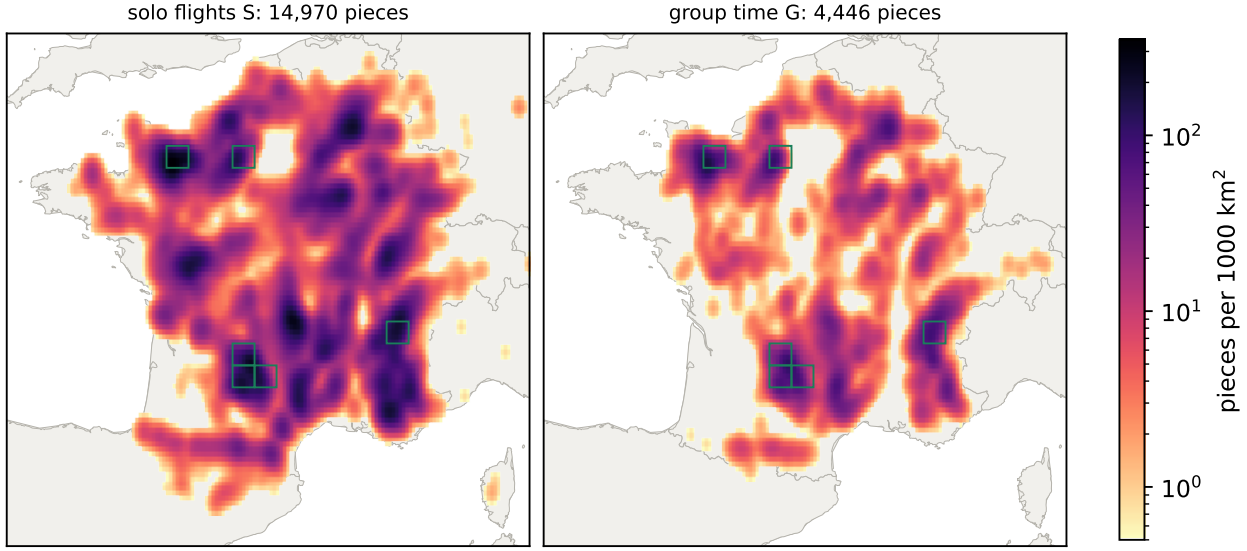


Figure 2: Density of the middles of the pieces of solo flights S (14 970 pieces, 14 192 flights) and of group time G (4 446 pieces, 3 342 flights), in pieces per 1000 km^2, on one colour scale. Squares: the zones of Section 5.

4 The raw difference

How do the three populations differ before any control?

The three populations progress almost alike: $H_{\text{loc}}(1000\text{ s})$ is 0.947 in G, 0.941 in A and 0.940 in S (Figure 3). They differ in the climbs and glides: group time spends more time climbing (42.1% against 38.7%), ends more searches in a climb and glides faster and flatter, with a glide ratio of 12.9 against 11.8, but climbs more slowly (1.08 against 1.13 m/s). The same flights alone are as straight as solo flights, spend the least time in climb, and sit between the two in the glide speed and the glide ratio.

4.1 Longer lags and the size of the group

How do the curves go over lags of hours, and with the number of pilots of the group?

A piece of at least 1250 s gives the lags up to 1000 s. A longer lag τ needs pieces of at least $\tau/0.8$ s, so a curve to a long lag is measured on the long pieces of its population only, and the population does not change along it. The populations of one panel of Figure 4 share the longest lag T, the longest at which every one of them keeps at least 100 flights, and are measured on the pieces of at least $T/0.8$ s (dashed); the thin lines are

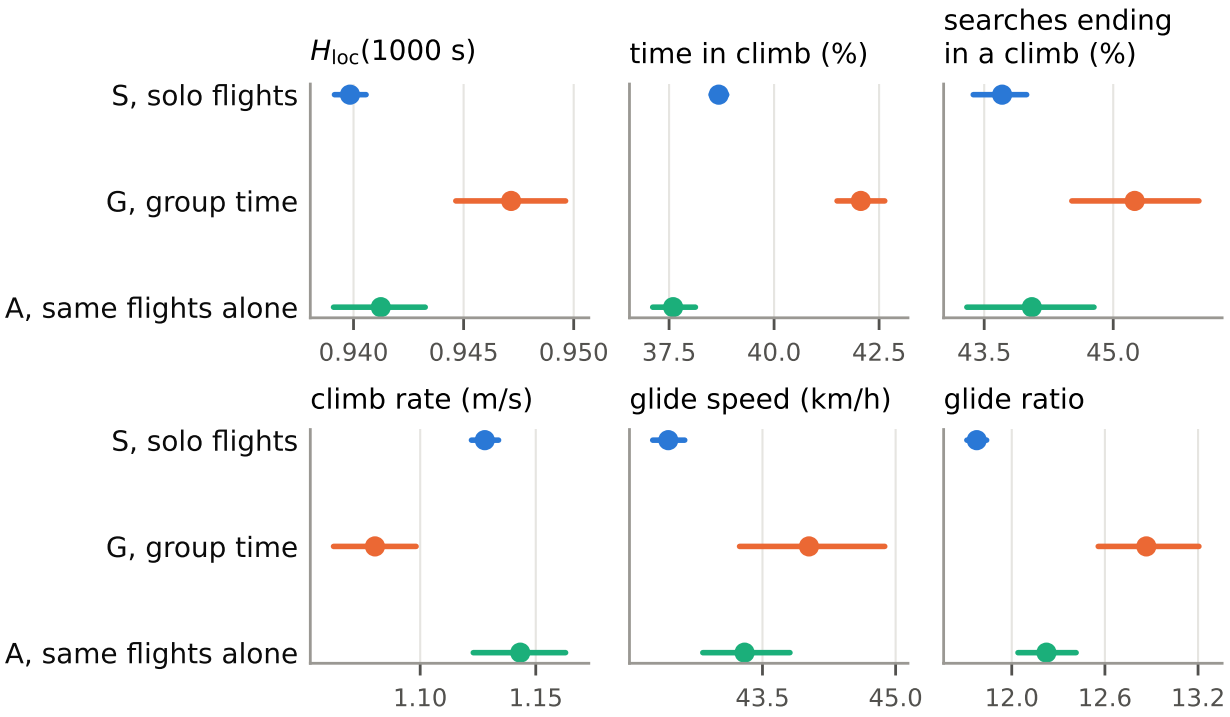


Figure 3: Pooled statistics of S, G and A with 5–95% bands. Exponents on 14 192, 3 342 and 2 577 flights, phase statistics on the flights of Table 1. Values in Table 2.

	solo flights S	group time G	same flights alone A
flights (TA-MSD)	14 192	3 342	2 577
H (10–1000 s)	0.946 [0.945, 0.946]	0.943 [0.941, 0.944]	0.946 [0.945, 0.947]
H_{loc}(1000 s)	0.940 [0.939, 0.941]	0.947 [0.945, 0.950]	0.941 [0.939, 0.943]
time in climb (%)	38.7 [38.5, 38.9]	42.1 [41.5, 42.6]	37.6 [37.1, 38.1]
searches ending in a climb (%)	43.7 [43.4, 44.0]	45.2 [44.5, 46.0]	44.1 [43.3, 44.8]
climb rate (m/s)	1.13 [1.12, 1.13]	1.08 [1.06, 1.10]	1.14 [1.12, 1.16]
glide speed (km/h)	42.4 [42.3, 42.6]	44.0 [43.2, 44.9]	43.3 [42.8, 43.8]
glide ratio	11.8 [11.7, 11.8]	12.9 [12.6, 13.2]	12.2 [12.0, 12.4]
cross-country speed (km/h)	28.8 [28.6, 28.9]	29.3 [28.6, 30.1]	29.9 [29.4, 30.4]

Table 2: Pooled statistics of the populations of Table 1, with 5–95% bands. The first row is the number of flights of the exponents; the other rows rest on the flights with phases (Table 1).

the curves of all the pieces up to 1000 s. The last panel gives the flights left at each lag, and Table 3 the flights of each curve. H_{loc} is read from MSD over the lags within ± 0.25 decades.

Pooled. On the long pieces the three populations stay close to a straight line up to lags of hours. At 5 080 s H_{loc} is 0.984 [0.978, 0.989] in G, against 0.977 in S and 0.978 in A; at 9 350 s it is 0.978 [0.965, 0.989] against 0.976 and 0.974, within the band of G, which rests on few flights (Table 3).

By size. Group time is straighter than solo flights at every size, and more so with more pilots: at 1 230 s H_{loc} is 0.945 in S, 0.951 [0.948, 0.953] for 2 pilots, 0.958, 0.958 and 0.962 for 3, 4 and 5–6 pilots, within each other's bands, and 0.986 [0.975, 0.996] for 7+ pilots.

Alone, by size. Alone, the flights of groups of every size are as straight as solo flights: at 4 150 s H_{loc} is 0.973 in S and from 0.973 to 0.977 for the flights of groups of 2 to 6 pilots; those of 7+ pilots have too few flights alone.

What we keep. On straight open flights group time and solo flights progress almost alike up to lags of hours, and group time is straighter only in the larger groups; the clear differences are in the climbs and glides.

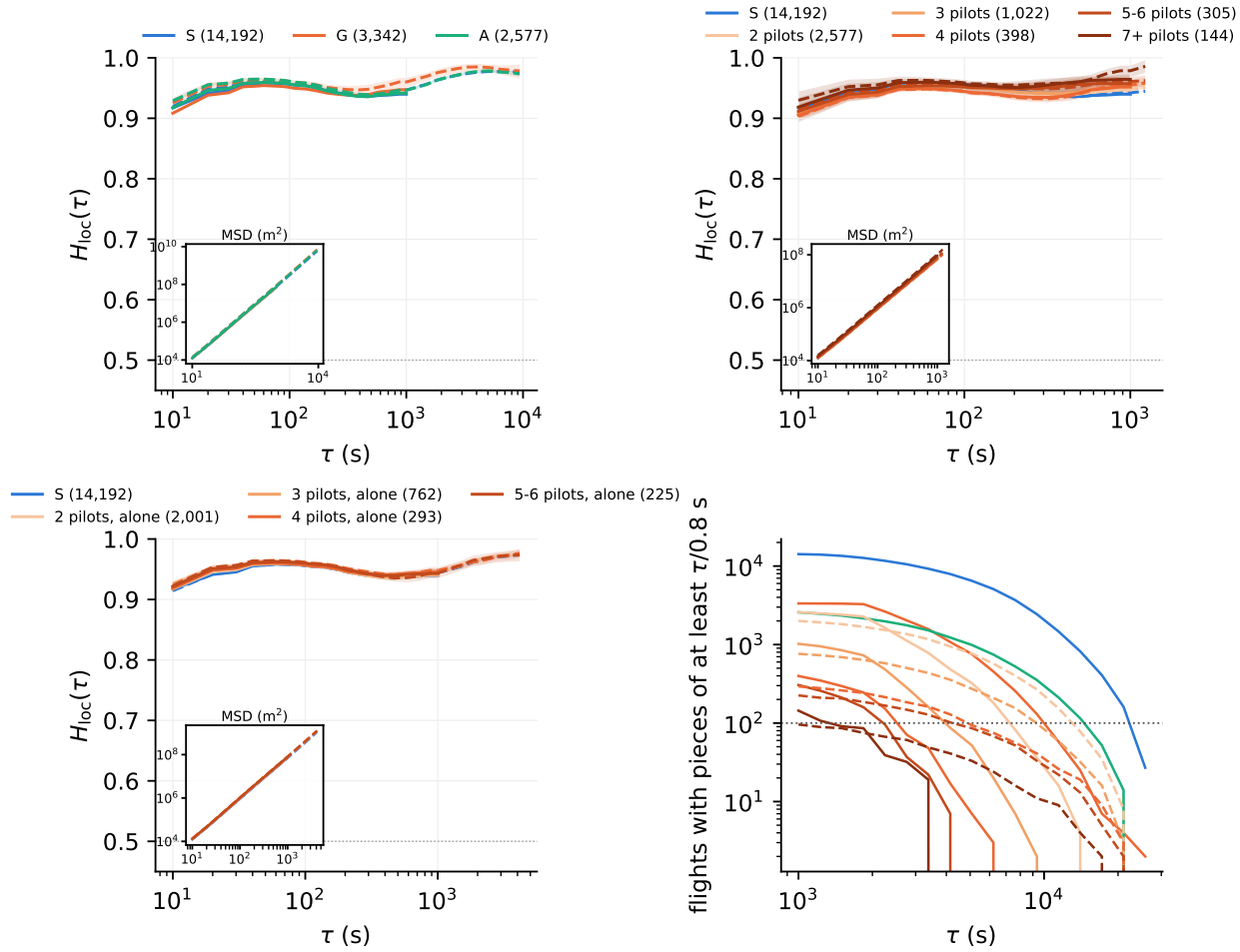


Figure 4: $H_{loc}(\tau)$ with an inset of MSD(τ) in m^2. Top left: S, G and A, lags up to 9 350 s. Top right: S and group time by the number of pilots of the group, up to 1 230 s. Bottom left: S and the same flights alone, by the size of the group the flight had, up to 4 150 s. Each population has two curves of its colour. Thin: all its pieces, up to 1000 s, with its flights in the legend. Dashed: only its pieces long enough for the longest lag of the panel, a subset of the thin ones, measured over the same population at every lag; 5–95% bands. Bottom right: flights with pieces of at least τ/0.8 s, one line per population (dashed for the flights alone); dotted, the minimum of 100 flights.

population	flights up to 1000 s	flights of the long series
pooled populations, lags up to 9 350 s		
solo flights S	14 192	2 436
group time G	3 342	130
same flights alone A	2 577	353
group time by the size of the group, lags up to 1 230 s		
solo flights S	14 192	13 988
G, 2 pilots	2 577	2 494
G, 3 pilots	1 022	950
G, 4 pilots	398	348
G, 5–6 pilots	305	258
G, 7+ pilots	144	107
the same flights alone, by size, lags up to 4 150 s		
solo flights S	14 192	7 916
A, flights of 2 pilots	2 001	945
A, flights of 3 pilots	762	356
A, flights of 4 pilots	293	129
A, flights of 5–6 pilots	225	102

Table 3: Flights of each curve of Figure 4: those of all the pieces, up to 1000 s, and those of the long series, with pieces of at least T/0.8 s for the longest lag T of the panel.

5 The zones with the most group flights

Do the three populations differ in the same way where groups are most frequent?

A zone is a cell of 50 km × 50 km that holds the middle of a piece, whichever the takeoff of the flight; the pilots of a zone need not take off from the same place. The 6 cells with the most flights that have group time in them are kept, and each is named by the takeoff most flown by its group flights (the cells are on a fixed grid; Figure 2). In each zone the pieces of S, G and A are those of Section 2, the measure is the same, and a population is drawn when it has at least 20 flights in the zone. Bands are 5–95% of bootstrap draws that resample the dependency blocks of the grouping.

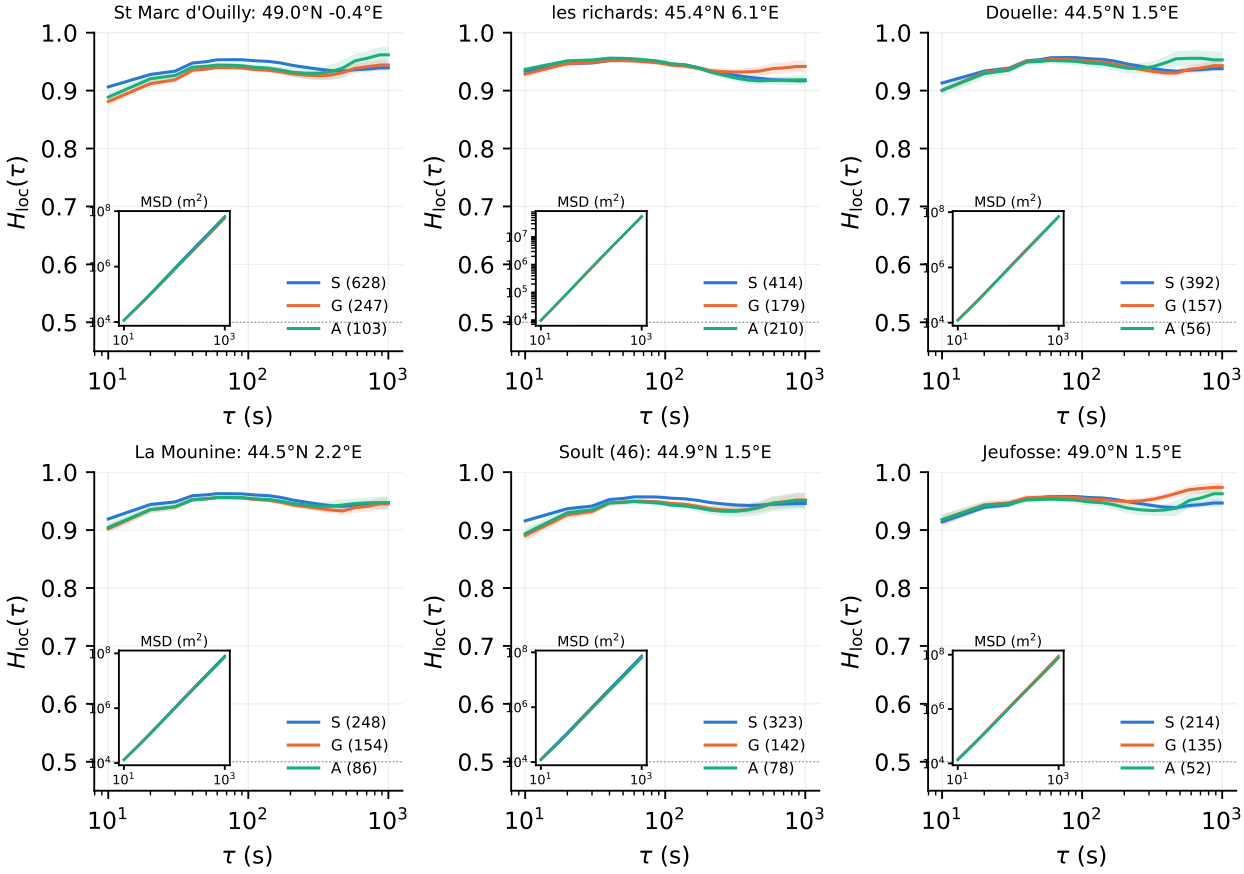


Figure 5: $H_{loc}(\tau)$ of S, G and A in the 6 zones, with the 5–95% band and, inset, MSD(τ) in m² (log–log). The number of flights of each population is in the legend and in Table 4.

In 3 of the 6 zones $H_{loc}(1000\text{ s})$ is highest in group time, then in the same flights alone, then in solo flights; in the others the same flights alone are as straight as group time or more (Figure 5, Table 4). Each zone holds few flights of group time.

zone	group flights	flights S	flights G	flights A	$H_{loc}(1000\text{ s})$ S	G	A
St Marc d'Ouilly (49.0°N, -0.4°E)	247	628	247	103	0.939	0.944	0.962
les richards (45.4°N, 6.1°E)	179	414	179	210	0.917	0.942	0.919
Douelle (44.5°N, 1.5°E)	157	392	157	56	0.938	0.943	0.953
La Mounine (44.5°N, 2.2°E)	154	248	154	86	0.946	0.946	0.947
Soult (46) (44.9°N, 1.5°E)	142	323	142	78	0.946	0.952	0.951
Jeufosse (49.0°N, 1.5°E)	135	214	135	52	0.947	0.974	0.963

Table 4: The zones, by the number of flights with group time in them. Flights of each population and $H_{loc}(1000\text{ s})$; the bands are in Figure 5.

What we keep. Zone by zone the three populations are close, and their order changes from one zone to the other.

6 The same route flown alone

Groups may form on a ridge or a convergence line where any pilot progresses straight, and the straightness would then belong to the route and not to the group. Does the gap remain when group time and its control fly the same route?

This test uses the declared open flights of Section 3, without the criterion on the opening: on the straight open flights too few pieces of group time have a control. The control is the same route flown alone, on another day. It differs from the route test of the general report (`vs-solo/general`) on three points: the control may come from any site and date other than that of the piece, and every piece of group time that has a control is used, not a sample; the control must also have had nobody within 10 km at the same time, since a flight with no contact in the grouping can still have flown near others; and a piece is compared with controls of the same wing level, with the results given for experienced and beginner pilots apart.

6.1 The control and the measure

Which flights are the same route flown alone, and how is the gap with group time measured?

Matching. The pieces of group time are those of Section 2. A segment k of a solo flight is a control of the piece g when all hold.

(i) **Route.** At least 80% of the points of g, one every 30 s, lie within 500 m of the track of the flight of k, which flew them in the same direction; k is the part of the track between its points nearest to the first and to the last of them, and lasts at least 1250 s.

(ii) **Date.** Other than that of g, at any site.

(iii) **Season and hour.** The middles of g and k are in months within one (circular) and in local hours within two.

(iv) **Wing level.** The same: beginners (A, B) or experienced (C, D, CCC) in the FFVL catalogue.

(v) **Solo.** The flight never had a contact of any length, which is stricter than S: a contact shorter than τ^* already excludes it.

(vi) **Isolation.** At every point of k, one every 30 s, no other flight of the archive, alone or in a group, paraglider or hang glider, was within 10 km within one minute (the index of the archive holds one position per minute).

All the matches are kept. 4 584 pieces have a match under the first five conditions and 3 224 after the isolation; they have 7 477 matches, a median of 1 per piece, from 1 276 solo flights of 765 pilots, against 1 536 pilots on the group side.

Comparison. In analogy with the zones of the general report, the stratum is the piece. The control of a piece is the mean of the MSD of its matches, and each piece weighs the same on both sides. With n pieces and m_g the matches of the piece g,

$$\mathrm{MSD}_{\mathrm{G}}(\tau) = \frac{1}{n}\sum_{g}\mathrm{MSD}_{g}(\tau), \qquad \mathrm{MSD}_{\mathrm{ctrl}}(\tau) = \frac{1}{n}\sum_{g}\frac{1}{m_g}\sum_{k\in g}\mathrm{MSD}_{k}(\tau), \tag{2}$$

and both sides are means of many MSDs before their logarithm. The gap is $\Delta H_{\mathrm{loc}}(\tau) = H^{\mathrm{G}}_{\mathrm{loc}}(\tau) - H^{\mathrm{ctrl}}_{\mathrm{loc}}(\tau)$, positive when group time progresses straighter. The baseline compares the same pieces with the solo flights of their site-day and hour, so that the two gaps refer to the same group time; the share of the baseline gap that the route explains is

$$1 - \frac{\Delta H^{\mathrm{route}}_{\mathrm{loc}}(1000\ \mathrm{s})}{\Delta H^{\mathrm{base}}_{\mathrm{loc}}(1000\ \mathrm{s})}. \tag{3}$$

Uncertainty. Bands are 5–95% of 1000 bootstrap draws that resample the pilots and the dependency blocks of the grouping, independently, and recompute every statistic; each difference has its own band. The thin lines of the figures are the range when one year at a time is left out. The baseline and the explained share are point estimates.

What we keep. The control is the same route, on another day, in the same season, hour and wing level, flown by a solo pilot with nobody within 10 km. One gap, ΔH_{loc}, is read at 1000 s and compared with the baseline of the same pieces.

6.2 What remains of the gap

How much of the straighter progress of groups is left once the route is the same?

Over a quarter of an hour, group time still progresses straighter than the same route flown alone: $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$ is $+0.016$ [$+0.004$, $+0.028$], against $+0.061$ for the same pieces against the solo flights of their site-day and hour (Table 5). Fixing the route lowers the gap by 74%, Eq. (3); its band stays away from zero. The long series, on the 1 949 pieces that last long enough for lags up to 1 970 s, has a larger gap, $+0.036$ [$+0.022$, $+0.052$] at 1000 s, with a band above zero (Figure 6).

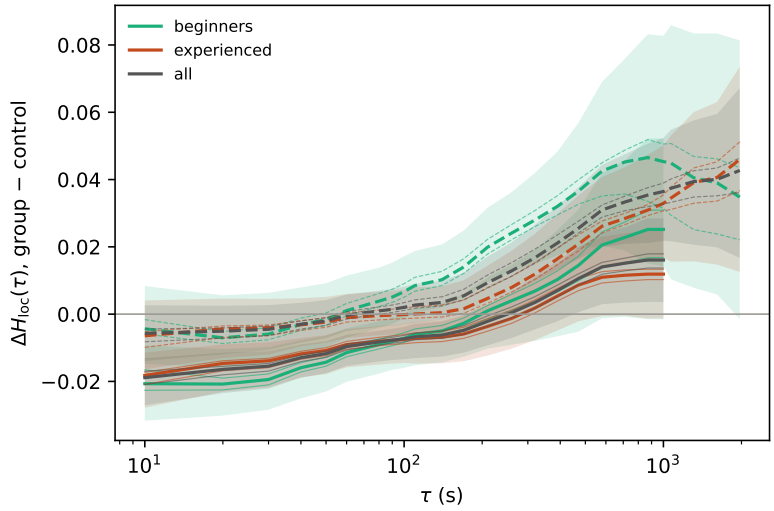


Figure 6: $\Delta H_{\mathrm{loc}}(\tau)$, group time minus the same route flown alone, for beginners (1 062 pieces), experienced pilots (2 162) and both (3 224). Continuous: all the pieces, lags up to 1000 s; dashed: only the pieces long enough for the longest lag, a subset of them, with lags beyond 1000 s. Bands 5–95%, thin lines the range when one year is left out.

The gap changes sign with the lag: at short lags, within one climb or glide, group time is less straight than the same route flown alone, and it is more straight from about 250 s on (Figure 6).

wing level	pieces	matches	against site-day and hour	against isolated, same route	explained
all	3 224	7 477	$+0.061$	$+0.016$ [$+0.004$, $+0.028$]	74%
experienced	2 162	5 419	$+0.054$	$+0.012$ [-0.001, $+0.025$]	78%
beginners	1 062	2 058	$+0.067$	$+0.025$ [-0.002, $+0.052$]	63%

Table 5: $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$ of group time against the solo flights of the same site-day and hour, and against the isolated solo flights on the same route, with the 5–95% band; the share of the first that the route explains, Eq. (3).

What we keep. The route, with the season, the hour and the wing level, explains about 74% of the long-lag gap between group and solo flight. The rest, $+0.016$ at 1000 s, is small and goes with flying together.

6.3 Experienced pilots and beginners

Does the route explain the same share for both wing levels?

Not quite (Table 5, Figure 7). For the 2 162 pieces of experienced pilots $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$ falls from $+0.054$ to $+0.012$ [-0.001, $+0.025$], 78% explained. For the 1 062 pieces of beginners it falls from $+0.067$ to $+0.025$ [-0.002, $+0.052$], 63% explained. The beginners' baseline gap is larger, their share lower and their band wider, since they have fewer pieces. Split by level, the gap that remains is not distinguished from zero.

What we keep. Both levels lose most of their gap on the same route, experienced pilots more than beginners, and neither remaining gap is distinguished from zero on its own.

6.4 Climbs and glides

Does a group climb and glide differently from the same route flown alone?

Table 6 compares the time in each phase and the statistics of climbs and glides of the pieces that have phases (loggers of 1 s) with the same statistics on their controls. No difference is beyond its band; the largest, a slower glide in group time, has a band that reaches zero.

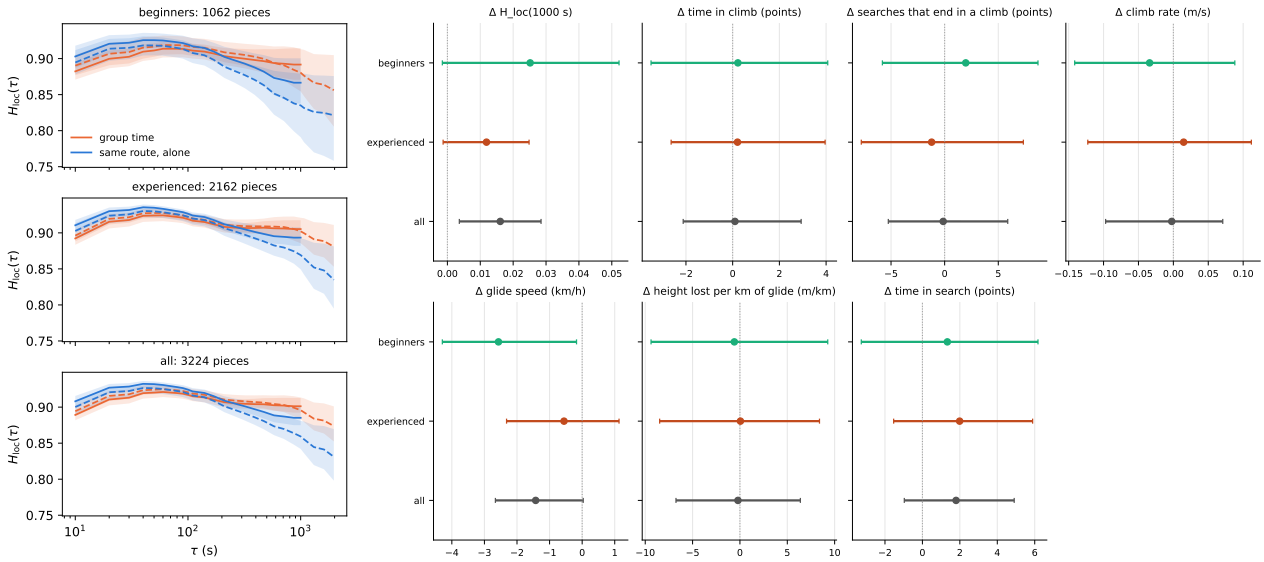


Figure 7: Left: $H_{\text{loc}}(\tau)$ of group time (orange) and of the same route flown alone (blue), by wing level, with the number of pieces in the panel titles. Continuous: all the pieces, up to 1000 s; dashed: only the pieces long enough for the longest lag, a subset of them, whose lags go further. Right: the difference group minus control of $H_{\text{loc}}(1000\ \text{s})$ and of the statistics of climbs and glides, with 5–95% bands, by wing level.

all levels	group time	same route alone	difference
pieces, segments	2 148	3 741	
time in glide (%)	56.5	58.4	-1.9 [-5.7, $+1.4$]
time in search (%)	14.8	13.0	$+1.8$ [-1.0, $+4.9$]
time in climb (%)	28.7	28.6	$+0.1$ [-2.1, $+2.9$]
searches ending in a climb (%)	49.3	49.5	-0.1 [-5.3, $+5.9$]
climbs per hour	7.54	7.27	$+0.27$ [-0.29, $+1.08$]
climb rate (m/s)	1.11	1.11	-0.00 [-0.10, $+0.07$]
glide speed (km/h)	28.9	30.3	-1.4 [-2.7, $+0.0$]
glide ratio	15.8	15.8	$+0.1$ [-1.7, $+1.7$]
height lost per km of glide (m/km)	63.2	63.4	-0.2 [-6.8, $+6.4$]

Table 6: Climbs and glides of group time and of the same route flown alone, both wing levels together, on 2 148 pieces of group time with phases and 3 741 matched segments; the difference group minus control with its 5–95% band. Times are shares of the time flown, in %, and their differences are in percentage points.

The right of Figure 7 gives the same differences for each level.

What we keep. On the same route a group does not climb or glide differently from the control beyond the sampling of the pieces.

7 Conclusions and limits

What can be concluded, and what cannot?

What we keep.

- Declared open flights fold back on themselves; we keep the straight ones, with an opening of at least 0.5, so that the shape of the flights does not pass for an effect of the group.
- On straight open flights group time (G, 3 342 flights), solo flights (S, 14 192) and the same flights alone (A, 2 577) progress almost alike, up to lags of hours; group time is straighter in the larger groups. Group time climbs more and glides faster and flatter. Zone by zone the order of the three populations changes.
- On the declared open flights, the route, with the season, the hour and the wing level, explains 74% of the gap $\Delta H_{\text{loc}}(1000\ \text{s})$ between group time and solo flights (78% for experienced pilots, 63% for beginners). What remains, $+0.016$ [$+0.004$, $+0.028$], is small; no statistic of climbs and glides differs from the control beyond its band.

Limits.

- The raw comparison fixes nothing: S and G differ in pilots, places, days and hours.
- The days differ: season and hour are matched, the weather of the day is not.
- The controls flew with nobody within 10 km. They may differ from the pilots who fly in groups, and this test has no selection contrast to measure it.
- Only the 3 224 pieces whose route was flown alone on another day enter; their routes are frequented ones. On the straight open flights too few pieces have a control, so the route test rests on the declared open flights.
- The bands measure the sampling of pilots and blocks. The radius and the cover of the route were varied only in the general report, on a sample, and the isolation distance not at all.

Most of the straighter progress of group time comes from the shape of the flights and from the route; what remains is small, and whether the group causes it stays open.